Introduction to Logic: Lecture 4

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November 10 2026

What this course is about (**reminder**)...

Logic: the study of reasoning, inference, and how one constructs correct arguments.

An argument in English

"You can access the Internet from campus only if **you are a linguistics major** or you are **not a freshman**. Mary **can access the Internet**. So, Mary is **a linguistics major** or **not a freshman**."

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We started translating into logic, today: inference

$$P_a \rightarrow (P_l \vee \neg P_f), \quad P_a \quad \therefore \quad P_l \vee \neg P_f$$

Tools we will learn along the way and **where we are**

Understanding and building logics requires a few mathematical tools:

$\{a, b, c\}$

Sets

linguistics majors

$R(a, b)$

Relations

loves, older than

$f : A \to B$

Functions

capital of

$\vdash$

Proofs

informal & formal

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first lecture	second lecture	second lecture	

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- Introduced classical propositional logic and its syntax and basic semantics, a new tool: **truth table**.

A new tool: truth tables

φ	ψ	$\varphi \wedge \psi$
1	1	**1**
1	0	**0**
0	1	**0**
0	0	**0**

left columns	the values of the *parts*
right column	the value of the *whole*
one row	one possible situation
all rows	*every* combination

A truth table lists a connective's function, one row per input. With 2 parts there are $2 \times 2 = 4$ rows; with n parts, 2^n (just like $|\mathcal{P}(X)| = 2^{|X|}$).

Proofs and logic: what you should know

Proofs

- ✓ definitions vs. theorems
- ✓ refute: *one* counterexample
- ✓ contradiction, construction

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Infinity

- ✓ infinite: bijection to a *part*
- ✓ countably infinite $=$ listable
- ✓ $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ countable, $\mathbb{R}$ is not

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Syntax of propositional logic

- ✓ letters, connectives, formulas
- ✓ trees and the main connective
- ✓ translate English $\leftrightarrow$ formulas

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Semantics

- ✓ every atom is $\mathbf{1}$ or $\mathbf{0}$
- ✓ truth tables for $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$
- ✓ $\rightarrow$ is 0 *only* for $1 \rightarrow 0$

Quick check: proofs and infinity

True or false?

1. one relation is enough to refute "every relation on $\{a, b\}$ is transitive"
2. no bijection $\{a, b, c\} \to \{a, b\}$ exists, so this alone shows $\{a, b, c\}$ is finite
3. the even numbers $\{0, 2, 4, \ldots\}$ are infinite
4. if A is countably infinite and $x \notin A$, then $A \cup \{x\}$ is countably infinite
5. every infinite subset of $\mathbb{R}$ is uncountable
6. there is a bijection $\mathbb{N} \to \mathbb{Z}$

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true

$\{\langle a, b\rangle, \langle b, a\rangle\}$: no $\langle a, a\rangle$

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   false
   check *every* proper subset

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   list x first: $x, a_0, a_1, \ldots$
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   $\mathbb{N} \subset \mathbb{R}$ is countable
6. there is a bijection $\mathbb{N} \to \mathbb{Z}$
   true
   the list $0, -1, 1, -2, 2, \ldots$

Quick check: syntax

True or false?

1. $\neg(P_a)$ is a formula

2. the main connective of $((P_a \vee \neg P_b) \rightarrow \neg P_c)$ is $\neg$

3. $((P_a \wedge P_b) \vee P_c)$ and $(P_a \wedge (P_b \vee P_c))$ have the same tree

4. $(P_a \rightarrow P_b)$ has the antecedent P_b

5. *it neither rains nor is the street wet* is $\neg(P_r \vee P_w)$

6. *it does not rain and the street is wet* is $\neg(P_r \wedge P_w)$

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   not (rain *or* wet)

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   false
   it is $(\neg P_r \wedge P_w)$

Quick check: semantics

True or false?

1. if P_a is 0, then $(P_a \rightarrow P_b)$ is 1, whatever P_b is
2. if P_a is 1 and P_b is 0, then $\neg(P_a \wedge \neg P_b)$ is 1
3. $(P_a \rightarrow P_b)$ and $(P_b \rightarrow P_a)$ always have the same value
4. $(P_a \vee \neg P_a)$ is 1 in every situation
5. a formula with 3 letters has a truth table with 6 rows
6. $(P_a \leftrightarrow P_b)$ is 1 when both letters are 0

Quick check: semantics

True or false?

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   a 0 antecedent makes $\rightarrow$ true

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   $\neg P_b = 1$, $\wedge$ gives 1, $\neg$ gives 0

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5. a formula with 3 letters has a truth table with 6 rows
   false
   $2 \times 2 \times 2 = 8$ rows
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   false
   $2 \times 2 \times 2 = 8$ rows
6. $(P_a \leftrightarrow P_b)$ is 1 when both letters are 0
   true
   same value: *neither*

Translation practice: hidden structure

Translate into propositional logic (examples from the reading)

1. *Berlin is east of Hamburg and Bremen.*
   P_h: B. is east of Hamburg, P_b: B. is east of Bremen
2. *Call me 'sweetie' once more and, no kidding, I'll step on your toe.*
   P_k: I'm not kidding, P_s: you call me 'sweetie', P_t: I step on your toe
3. *I will only play salsa, if you give me earplugs.*
   P_p: I play salsa, P_e: you give me earplugs
4. *If drawing a red card means that I lost, I lost.*
   P_r: I drew a red card, P_l: I lost

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1. *Berlin is east of Hamburg and Bremen.* $(P_h \wedge P_b)$
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   one sentence, two propositions

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2. *Call me 'sweetie' once more and, no kidding, I'll step on your toe.*
   P_k: I'm not kidding, P_s: you call me 'sweetie', P_t: I step on your toe

   $(P_k \wedge (P_s \rightarrow P_t))$
   this *and* means *if*

3. *I will only play salsa, if you give me earplugs.*
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3. *I will only play salsa, if you give me earplugs.*
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   $(P_p \rightarrow P_e)$
   earplugs are *necessary*, not sufficient

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   P_r: I drew a red card, P_l: I lost
   $((P_r \rightarrow P_l) \rightarrow P_l)$
   an *if* inside an *if*

Translation practice: tricky words

Translate into propositional logic

P_r: it rains, P_w: the street is wet, P_c: class is cancelled, P_t: the teacher comes

1. *It rains, but the street is dry.*

2. *It neither rains nor is the street wet.*

3. *You can't have both rain and a dry street.*

4. *Class is cancelled only if it rains.*

5. *Class is cancelled unless the teacher comes.*

6. *Either it rains or class is cancelled (not both).*

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P_r: it rains, P_w: the street is wet, P_c: class is cancelled, P_t: the teacher comes

1. *It rains, but the street is dry.* $(P_r \wedge \neg P_w)$
   but is $\wedge$
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2. *It neither rains nor is the street wet.* $\neg(P_r \vee P_w)$
   or: $(\neg P_r \wedge \neg P_w)$
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   not both: $\neg$ outside
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   not both: $\neg$ outside
4. *Class is cancelled only if it rains.* — $(P_c \rightarrow P_r)$
   only if points the other way!
5. *Class is cancelled unless the teacher comes.*
6. *Either it rains or class is cancelled (not both).*

Translation practice: tricky words

Translate into propositional logic

P_r: it rains, P_w: the street is wet, P_c: class is cancelled, P_t: the teacher comes

1. *It rains, but the street is dry.* — $(P_r \wedge \neg P_w)$
   but is $\wedge$
2. *It neither rains nor is the street wet.* — $\neg(P_r \vee P_w)$
   or: $(\neg P_r \wedge \neg P_w)$
3. *You can't have both rain and a dry street.* — $\neg(P_r \wedge \neg P_w)$
   not both: $\neg$ outside
4. *Class is cancelled only if it rains.* — $(P_c \rightarrow P_r)$
   only if points the other way!
5. *Class is cancelled unless the teacher comes.* — $(\neg P_t \rightarrow P_c)$
   unless = *if not*
6. *Either it rains or class is cancelled (not both).*

Translation practice: tricky words

Translate into propositional logic

P_r: it rains, P_w: the street is wet, P_c: class is cancelled, P_t: the teacher comes

1. *It rains, but the street is dry.* — $(P_r \wedge \neg P_w)$
   but is $\wedge$
2. *It neither rains nor is the street wet.* — $\neg(P_r \vee P_w)$
   or: $(\neg P_r \wedge \neg P_w)$
3. *You can't have both rain and a dry street.* — $\neg(P_r \wedge \neg P_w)$
   not both: $\neg$ outside
4. *Class is cancelled only if it rains.* — $(P_c \rightarrow P_r)$
   only if points the other way!
5. *Class is cancelled unless the teacher comes.* — $(\neg P_t \rightarrow P_c)$
   unless = *if not*
6. *Either it rains or class is cancelled (not both).* — $((P_r \vee P_c) \wedge \neg(P_r \wedge P_c))$
   exclusive or

Questions?

Truth tables for complex formulas

What we covered: truth tables for connectives

One truth table for all five connectives

φ	ψ	$\neg\varphi$	$\varphi\wedge\psi$	$\varphi\vee\psi$	$\varphi\rightarrow\psi$	$\varphi\leftrightarrow\psi$
1	1	**0**	**1**	**1**	**1**	**1**
1	0	**0**	**0**	**1**	**0**	**0**
0	1	**1**	**0**	**1**	**1**	**0**
0	0	**1**	**0**	**0**	**1**	**1**
		not: flips φ	**and**: *both* are 1	**or**: *at least one* is 1	**if**: 0 only in row 2	**iff**: *same* value

Sound familiar? The same words defined the set operations

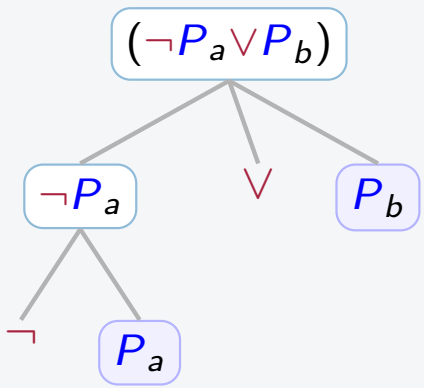
$X \cap Y = \{z \mid z \in X \textbf{ and } z \in Y\}$ $\quad \overline{X} = \{z \in U \mid \textbf{not } z \in X\}$

$X \cup Y = \{z \mid z \in X \textbf{ or } z \in Y\}$

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

The tree



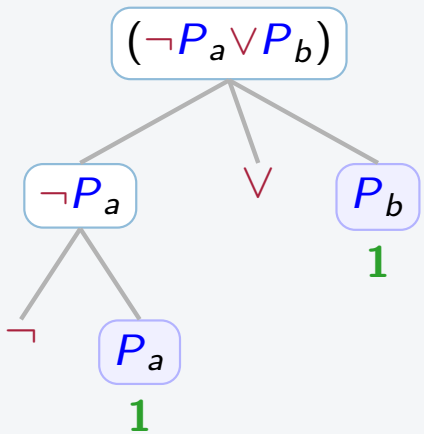
The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$
s_1	1	1	
s_2	1	0	
s_3	0	1	
s_4	0	0	

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

Row s_1



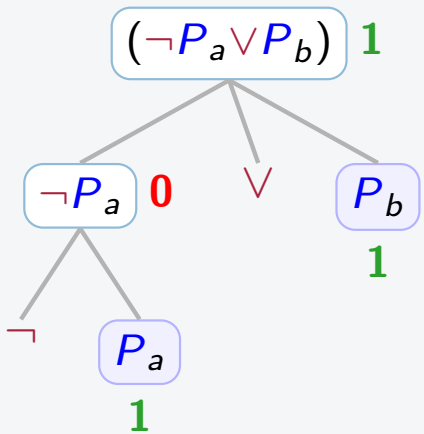
The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$
s_1	1	1	$\leftarrow$
s_2	1	0	
s_3	0	1	
s_4	0	0	

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

Row s_1



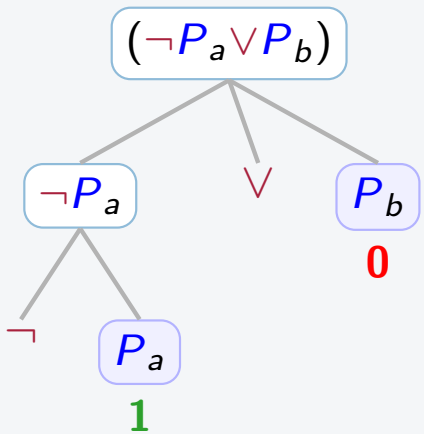
The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$	
s_1	1	1	1	$\leftarrow$
s_2	1	0		
s_3	0	1		
s_4	0	0		

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

Row s_2



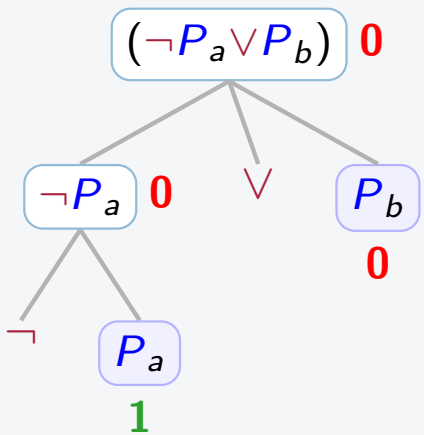
The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$
s_1	1	1	1
s_2	1	0	$\leftarrow$
s_3	0	1	
s_4	0	0	

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

Row s_2



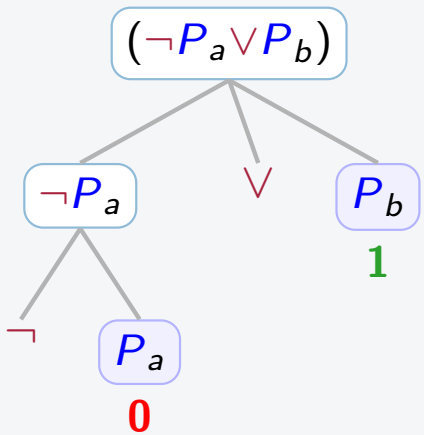
The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$	
s_1	1	1	**1**	
s_2	1	0	**0**	$\leftarrow$
s_3	0	1		
s_4	0	0		

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

Row s_3



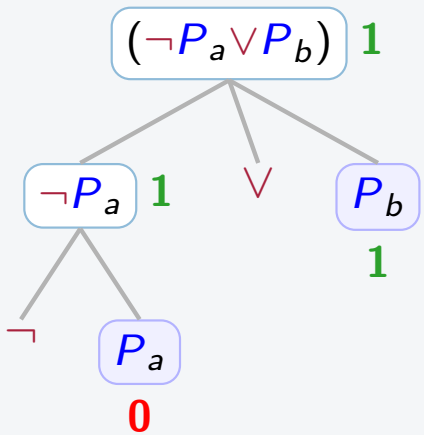
The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$
s_1	1	1	**1**
s_2	1	0	**0**
s_3	0	1	←
s_4	0	0	

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

Row s_3



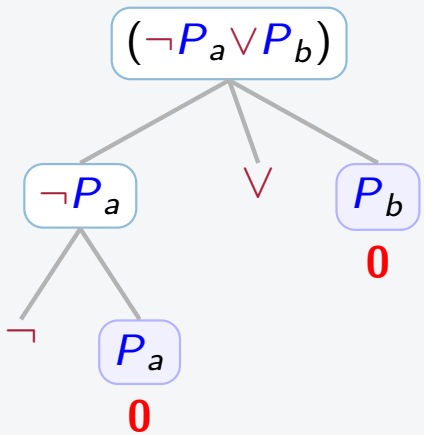
The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$	
s_1	1	1	1	
s_2	1	0	0	
s_3	0	1	1	←
s_4	0	0		

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

Row s_4



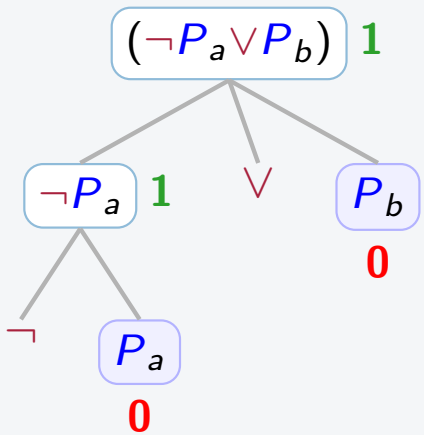
The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$
s_1	1	1	**1**
s_2	1	0	**0**
s_3	0	1	**1**
s_4	0	0	←

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

Row s_4



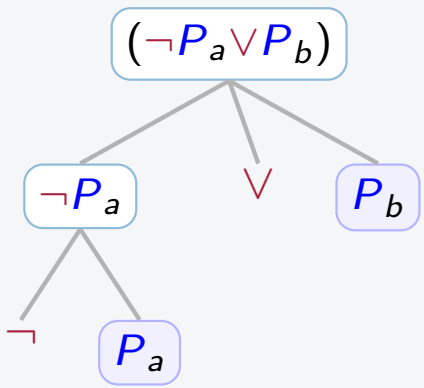
The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$	
s_1	1	1	1	
s_2	1	0	0	
s_3	0	1	1	
s_4	0	0	1	$\leftarrow$

Computing a truth table: one row at a time

- For each row: **1.** put the row's values at the *leaves*; **2.** work *up* the tree, one connective at a time.

The tree



The truth table

	P_a	P_b	$(\neg P_a \vee P_b)$
s_1	1	1	**1**
s_2	1	0	**0**
s_3	0	1	**1**
s_4	0	0	**1**

Every row is computed the **same way**: purely mechanical, a computer could do it.

the formula is 0 only in s_2: does that remind you of another connective?

Two formulas, the same column

Put both formulas side by side

	P_a	P_b	$(\neg P_a \vee P_b)$		
s_1	1	1	**1**		
s_2	1	0	**0**		
s_3	0	1	**1**		
s_4	0	0	**1**		

Two formulas, the same column

Put both formulas side by side

	P_a	P_b	$(\neg P_a \vee P_b)$	$(P_a \rightarrow P_b)$	
s_1	1	1	**1**	**1**	
s_2	1	0	**0**	**0**	
s_3	0	1	**1**	**1**	
s_4	0	0	**1**	**1**	

Two formulas, the same column

Put both formulas side by side

	P_a	P_b	$(\neg P_a \vee P_b)$	$(P_a \rightarrow P_b)$	same?
s_1	1	1	**1**	**1**	✓
s_2	1	0	**0**	**0**	✓
s_3	0	1	**1**	**1**	✓
s_4	0	0	**1**	**1**	✓

Two formulas, the same column

Put both formulas side by side

	P_a	P_b	$(\neg P_a \vee P_b)$	$(P_a \rightarrow P_b)$	same?
s_1	1	1	**1**	**1**	✓
s_2	1	0	**0**	**0**	✓
s_3	0	1	**1**	**1**	✓
s_4	0	0	**1**	**1**	✓

with P_a: *it rains*, P_b: *the street is wet*

it doesn't rain, or the street is wet vs. *if it rains, then the street is wet*

different words, true in the same situations

Two formulas, the same column

Put both formulas side by side

	P_a	P_b	$(\neg P_a \vee P_b)$	$(P_a \rightarrow P_b)$	same?
s_1	1	1	**1**	**1**	✓
s_2	1	0	**0**	**0**	✓
s_3	0	1	**1**	**1**	✓
s_4	0	0	**1**	**1**	✓

with P_a: *it rains*, P_b: *the street is wet*

it doesn't rain, or the street is wet vs. *if it rains, then the street is wet*

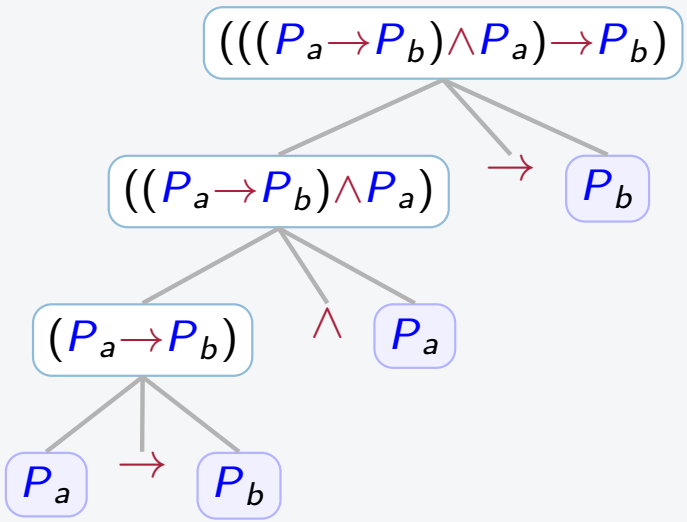
different words, true in the same situations

Two *different* formulas, the *same* value in **every** row: they say the same thing, this is a **logical equivalence**.

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

The tree



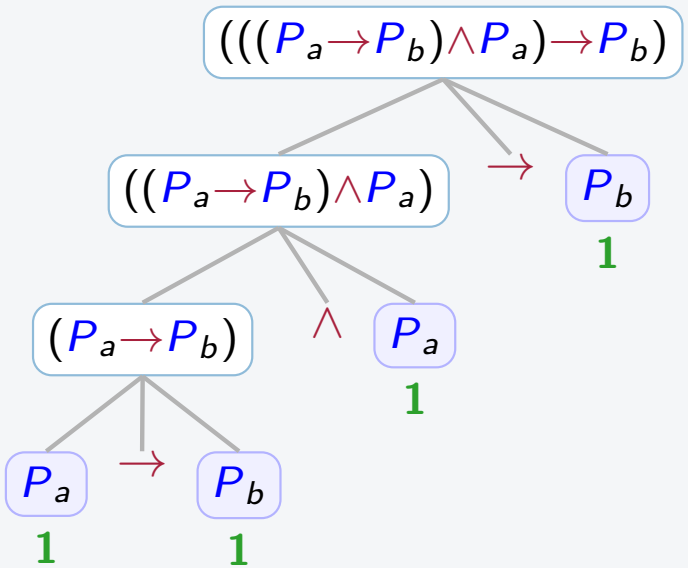
The truth table

	P_a	P_b	formula
s_1	1	1	
s_2	1	0	
s_3	0	1	
s_4	0	0	

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

Row s_1



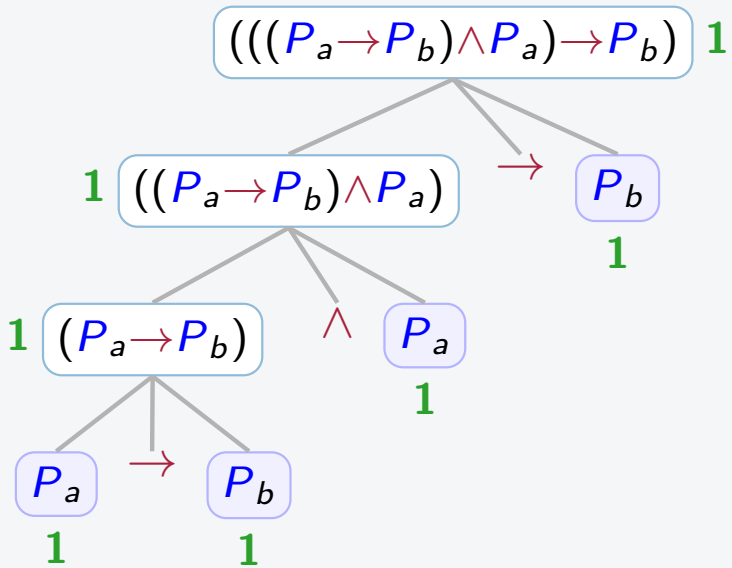
The truth table

	P_a	P_b	formula
s_1	1	1	
s_2	1	0	
s_3	0	1	
s_4	0	0	

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

Row s_1



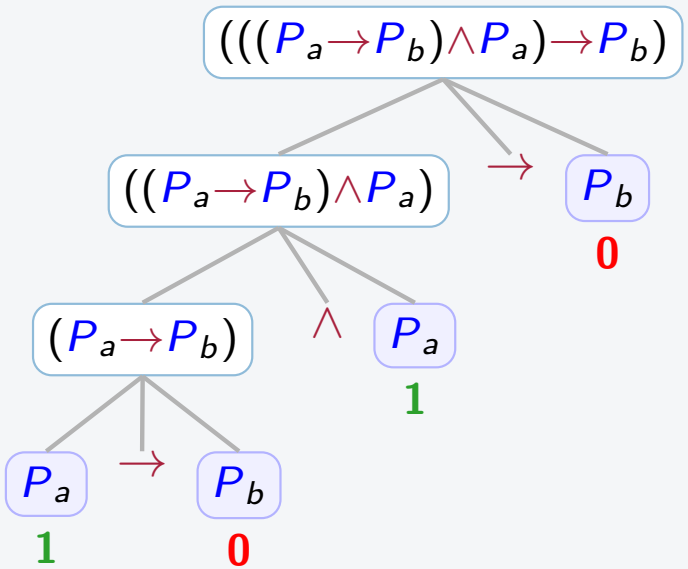
The truth table

	P_a	P_b	formula
s_1	1	1	**1**
s_2	1	0	
s_3	0	1	
s_4	0	0	

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

Row s_2



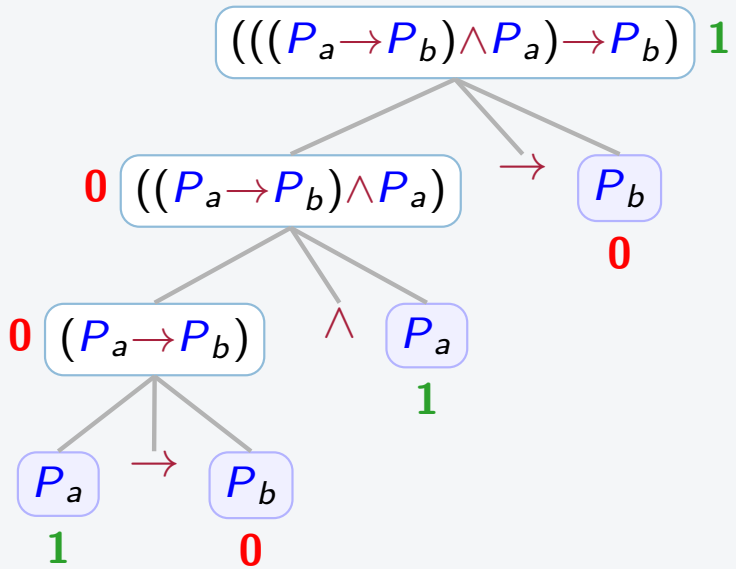
The truth table

	P_a	P_b	formula
s_1	1	1	1
s_2	1	0	
s_3	0	1	
s_4	0	0	

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

Row s_2



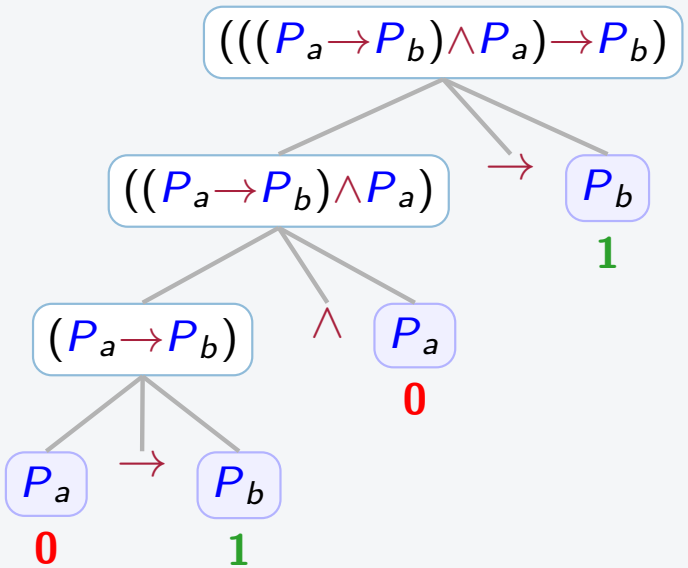
The truth table

	P_a	P_b	formula
s_1	1	1	1
s_2	1	0	1
s_3	0	1	
s_4	0	0	

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

Row s_3



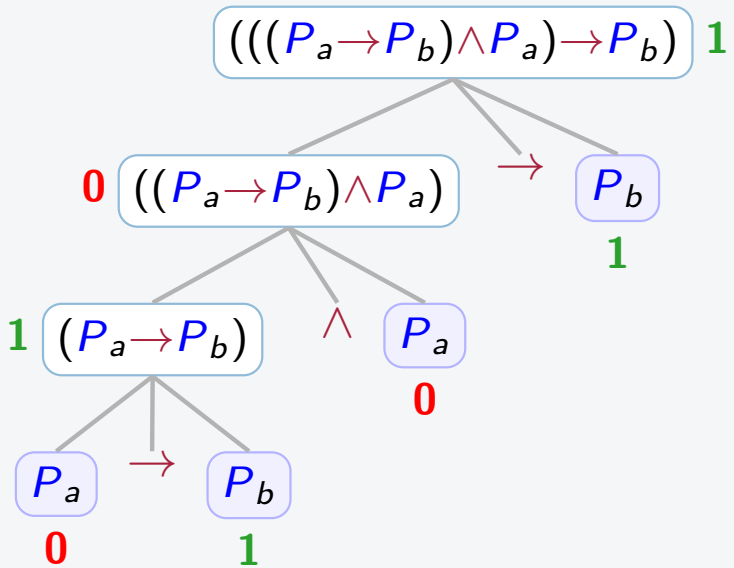
The truth table

	P_a	P_b	formula
s_1	1	1	**1**
s_2	1	0	**1**
s_3	0	1	
s_4	0	0	

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

Row s_3



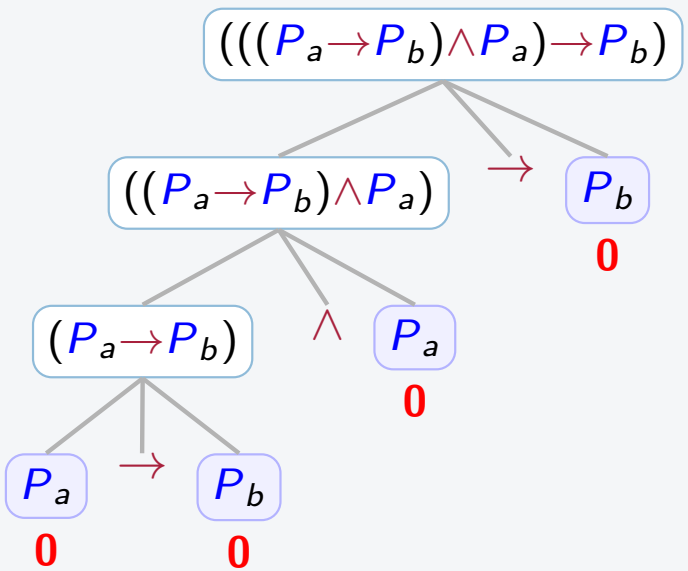
The truth table

	P_a	P_b	formula
s_1	1	1	**1**
s_2	1	0	**1**
s_3	0	1	**1**
s_4	0	0	

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

Row s_4



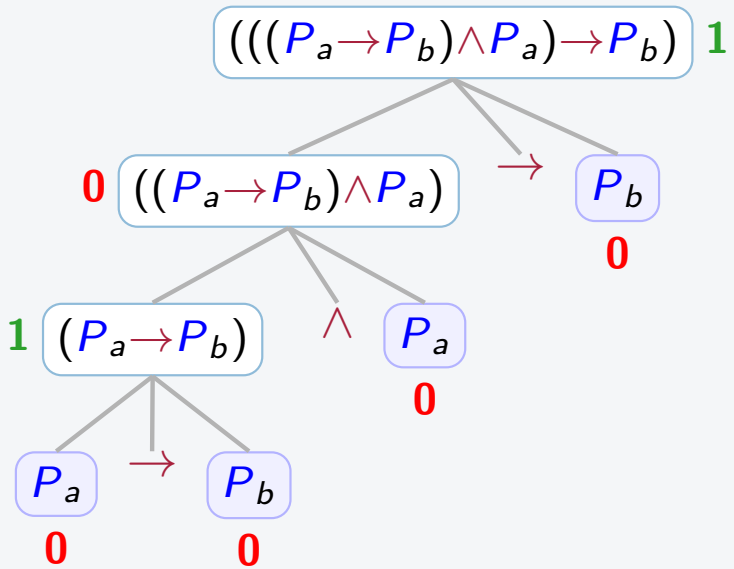
The truth table

	P_a	P_b	formula
s_1	1	1	**1**
s_2	1	0	**1**
s_3	0	1	**1**
s_4	0	0	

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

Row s_4



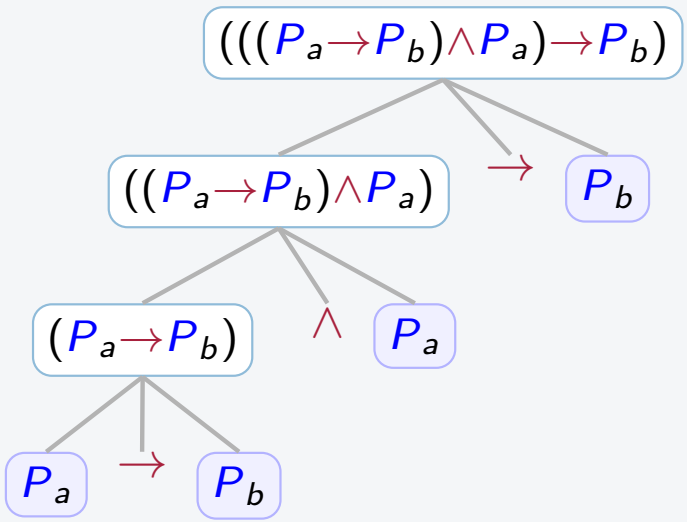
The truth table

	P_a	P_b	formula
s_1	1	1	**1**
s_2	1	0	**1**
s_3	0	1	**1**
s_4	0	0	**1**

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

The tree



The truth table

	P_a	P_b	formula
s_1	1	1	**1**
s_2	1	0	**1**
s_3	0	1	**1**
s_4	0	0	**1**

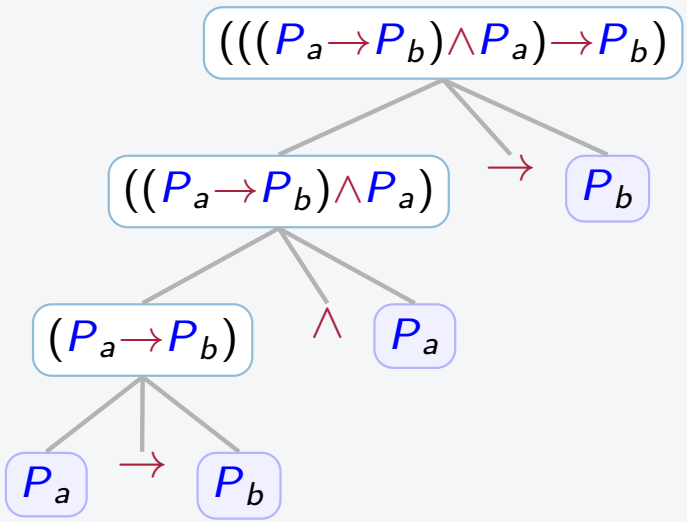
This formula is **1 in every row**! It says: *if* $P_a \to P_b$ *and* P_a, *then* P_b.

the shape of our argument about Mary: such formulas get a name today

A bigger example

- Same procedure, deeper tree. Note: P_a and P_b appear *twice*, and get the *same* value in both places.

The tree



The truth table

	P_a	P_b	formula
s_1	1	1	**1**
s_2	1	0	**1**
s_3	0	1	**1**
s_4	0	0	**1**

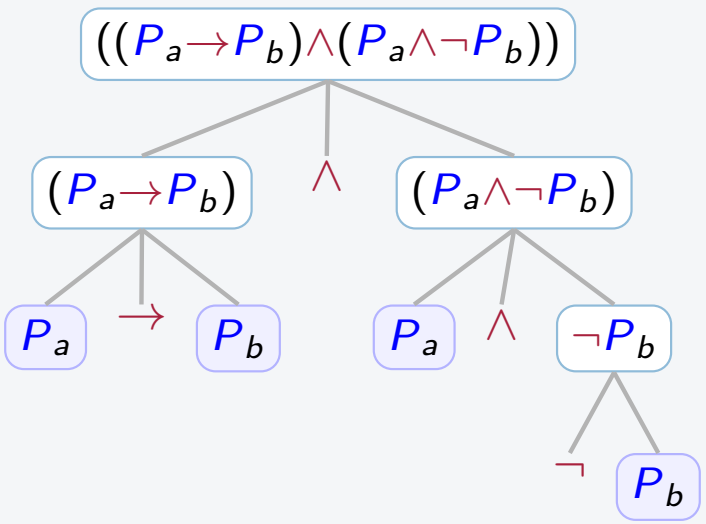
A formula that is **1 in every row** is called a **tautology**.

true by its *form* alone, whatever P_a and P_b mean

One more example

- Same procedure again. Watch the root: what happens in *every* row?

The tree



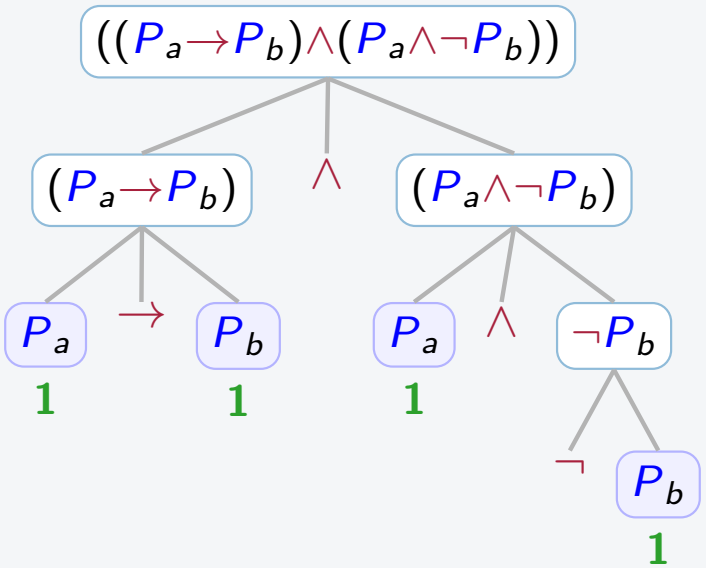
The truth table

	P_a	P_b	formula
s_1	1	1	
s_2	1	0	
s_3	0	1	
s_4	0	0	

One more example

- Same procedure again. Watch the root: what happens in *every* row?

Row s_1



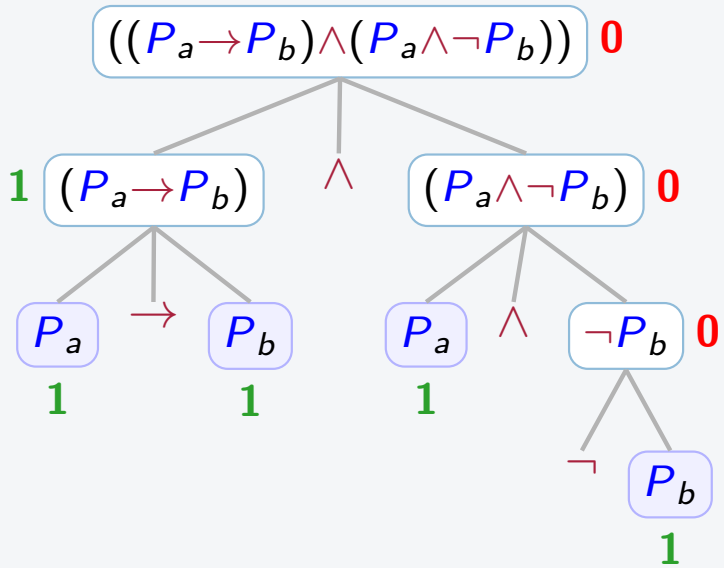
The truth table

	P_a	P_b	formula
s_1	1	1	
s_2	1	0	
s_3	0	1	
s_4	0	0	

One more example

- Same procedure again. Watch the root: what happens in *every* row?

Row s_1



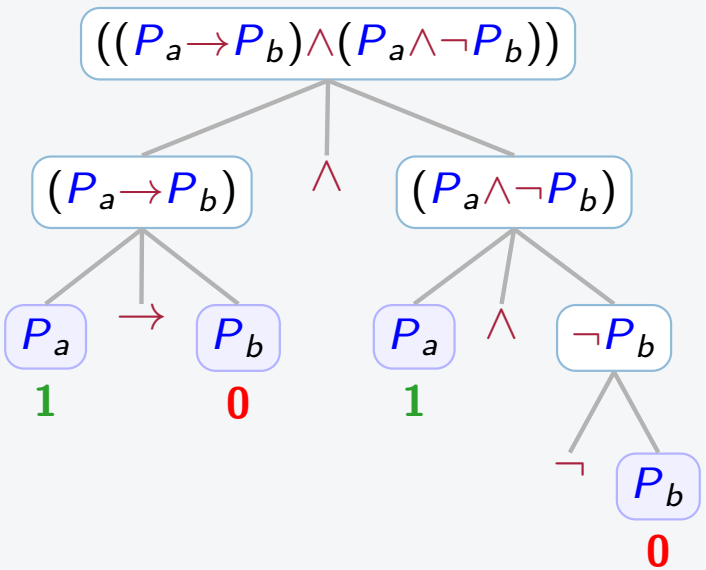
The truth table

	P_a	P_b	formula
s_1	1	1	**0**
s_2	1	0	
s_3	0	1	
s_4	0	0	

One more example

- Same procedure again. Watch the root: what happens in *every* row?

Row s_2



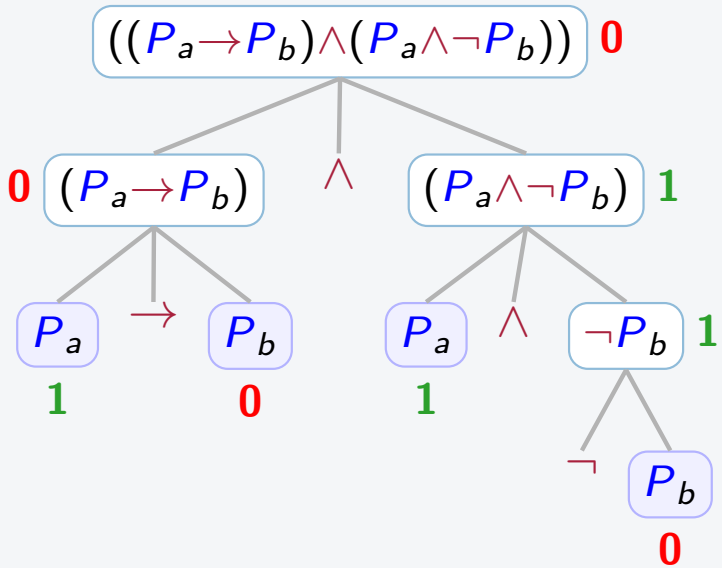
The truth table

	P_a	P_b	formula
s_1	1	1	**0**
s_2	1	0	
s_3	0	1	
s_4	0	0	

One more example

- Same procedure again. Watch the root: what happens in *every* row?

Row s_2



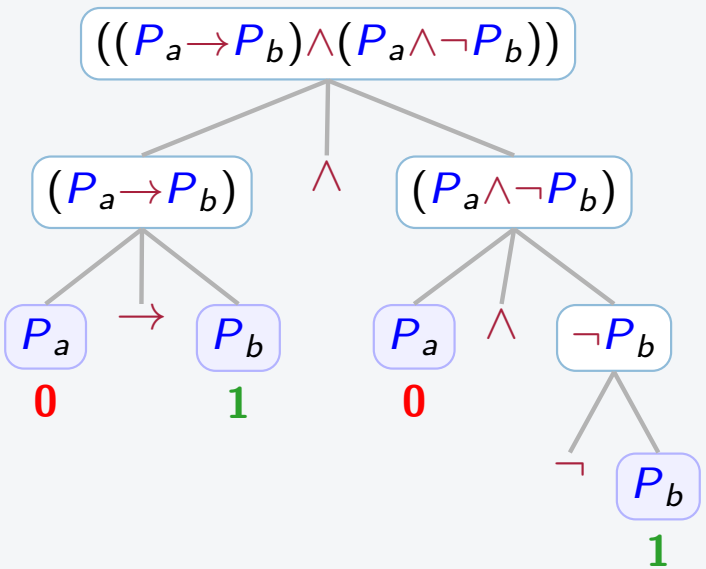
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s_3	0	1	
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Row s_3



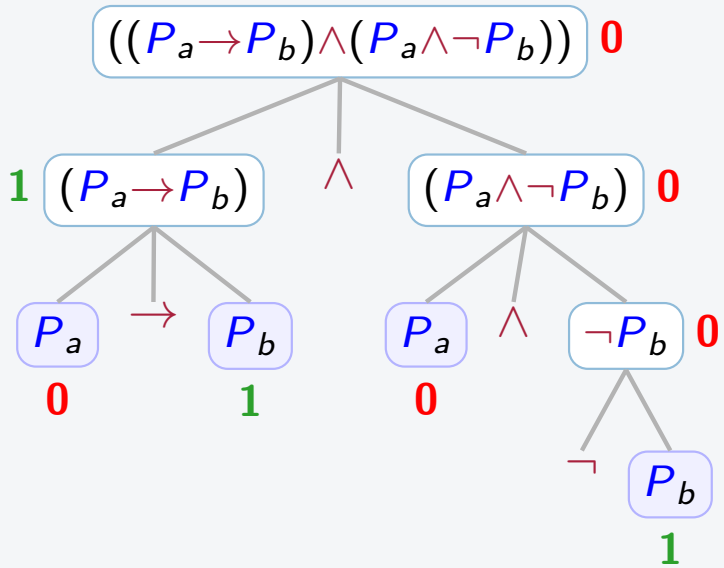
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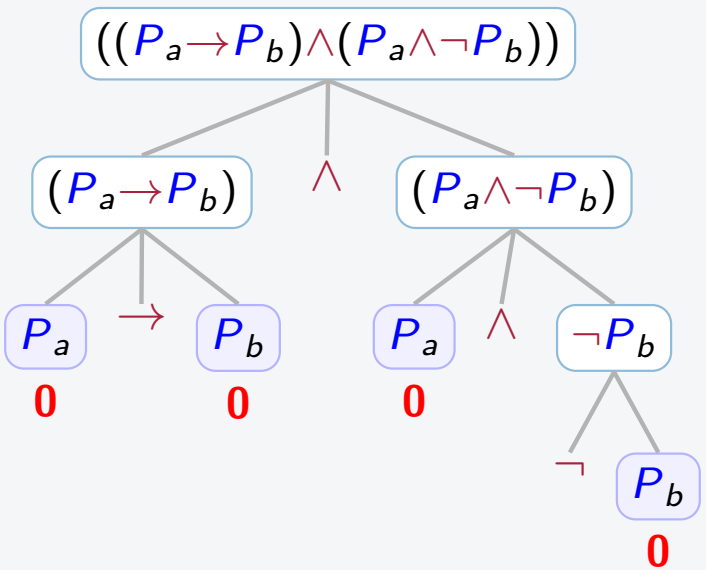
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Row s_4



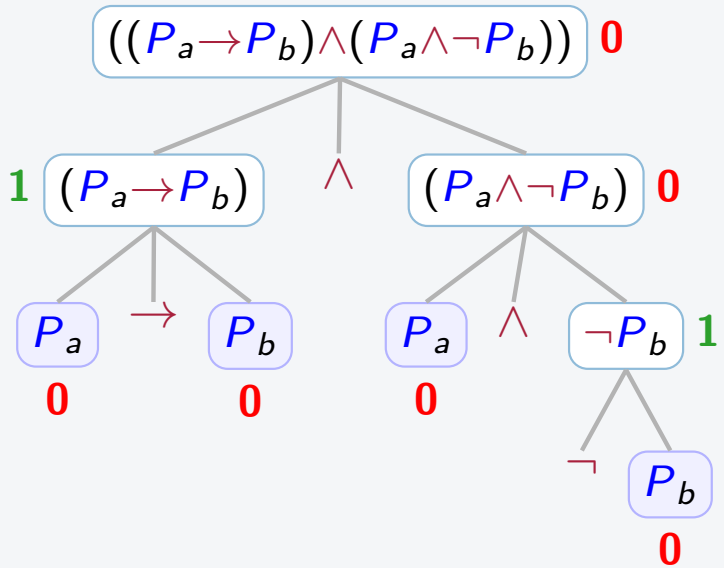
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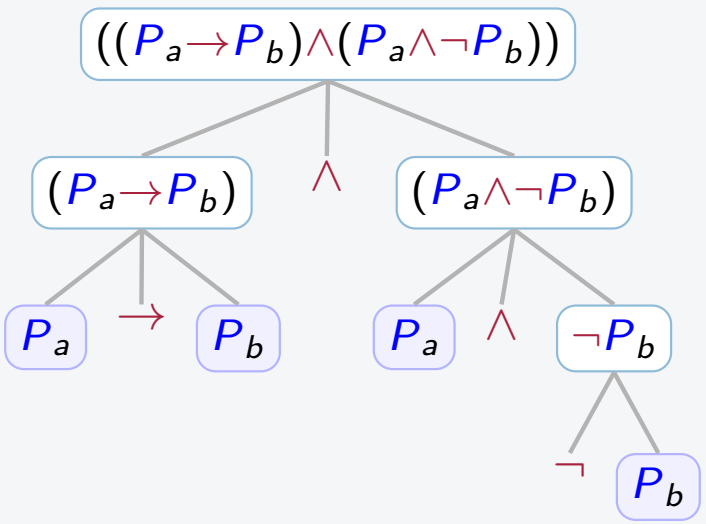
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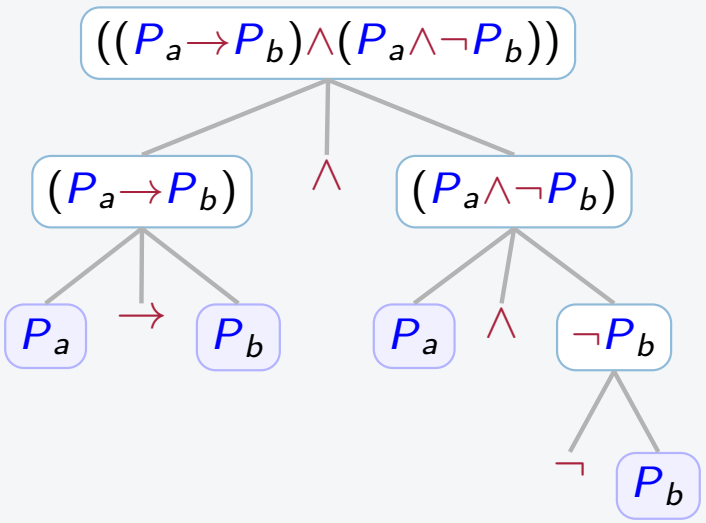
0 in every row! It says: $P_a \to P_b$, *and* P_a, *and not* P_b.

the shape of Mary's premises, plus the *opposite* of the conclusion

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A formula that is **0 in every row** is called **unsatisfiable**.

no situation can make it true (also called a *contradiction*)

The meaning of a formula: a set of situations

Definition: the meaning of a formula

Let $S = \{s_1, s_2, \ldots\}$ be the set of **situations**, one for each row of the truth table (n letters: $|S| = 2^n$). The **meaning** of a formula φ is the set of situations in which φ is true:

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$V(p), V(q), \ldots$		arbitrary
$V(\neg\phi) = 1$	iff	$V(\phi) = 0$
$V(\phi \wedge \psi) = 1$	iff	$V(\phi) = 1$ and $V(\psi) = 1$
$V(\phi \vee \psi) = 1$	iff	$V(\phi) = 1$ or $V(\psi) = 1$
$V(\phi \rightarrow \psi) = 0$	iff	$V(\phi) = 1$ and $V(\psi) = 0$
$V(\phi \leftrightarrow \psi) = 1$	iff	$V(\phi) = V(\psi)$

Read

s_1				
s_2				$s_4\}$
s_3	0	1	0	1
s_4	0	0	0	1

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   leaves first, then *up* the tree

   row by row, mechanically

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   $[\![\varphi]\!] = \{s \in S \mid \varphi \text{ is 1 in } s\}$

   where φ is true

Quick check: truth tables and meanings

$$s_1 = \underbrace{1\,1}_{P_a\,P_b} \qquad s_2 = 1\,0 \qquad s_3 = 0\,1 \qquad s_4 = 0\,0$$

each situation, written as the values of P_a and P_b in that row

True or false?

1. $(P_a \wedge P_b)$ and $(P_b \wedge P_a)$ have the same column
2. $(P_a \wedge \neg P_a)$ is a tautology
3. $[\![P_b]\!] = \{s_1, s_3\}$
4. $[\![\neg P_a]\!] = \{s_1, s_2\}$
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   true
   $\neg$ flips every row: the complement

The core semantic notions in logic

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$\varphi \equiv \psi$

same **meaning**?

1111

always true?

0000

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$\varphi \models \psi$

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φ and ψ are **logically equivalent**, written $\varphi \equiv \psi$, iff $[\![\varphi]\!] = [\![\psi]\!]$

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Example

$$[\![(\neg P_a \vee P_b)]\!] = \{s_1, s_3, s_4\} = [\![(P_a \rightarrow P_b)]\!]$$

so $(\neg P_a \vee P_b) \equiv (P_a \rightarrow P_b)$

Important rules for computing logical equivalences

For *any* formulas φ, ψ, χ

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De Morgan	$\neg(\varphi \wedge \psi) \equiv (\neg\varphi \vee \neg\psi)$
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Important rules for computing logical equivalences

For *any* formulas φ, ψ, χ

double negation	$\neg\neg\varphi \equiv \varphi$
commutativity	$(\varphi \wedge \psi) \equiv (\psi \wedge \varphi), \quad (\varphi \vee \psi) \equiv (\psi \vee \varphi)$
associativity	$((\varphi \wedge \psi) \wedge \chi) \equiv (\varphi \wedge (\psi \wedge \chi))$ same for $\vee$
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Each can be checked with a truth table. De Morgan for sets (lecture 1): $\overline{X \cap Y} = \overline{X} \cup \overline{Y}$

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Study these, go through carefully!

The core semantic notions in logic

$\varphi \equiv \psi$	1111	0000	$\varphi \models \psi$
same **meaning**?	**always** true?	**never** true?	what **follows**?

Definition: tautology

φ is a **tautology** iff $[\![\varphi]\!] = S$

true in *every* possible situation or world

e.g., $(P_a \vee \neg P_a)$

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true in *no* situation (a *contradiction*)

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In between

φ is **satisfiable** iff $[\![\varphi]\!] \neq \emptyset$: true in *some* situation e.g., P_a, $(P_a \wedge P_b)$

The core semantic notions in logic: **entailment**

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same **meaning**?	**always** true?	**never** true?	what **follows**?

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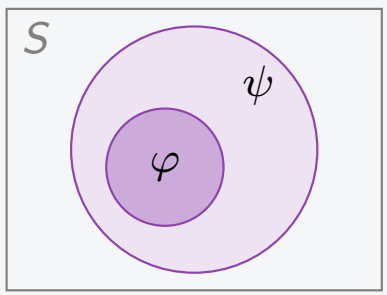
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In a picture



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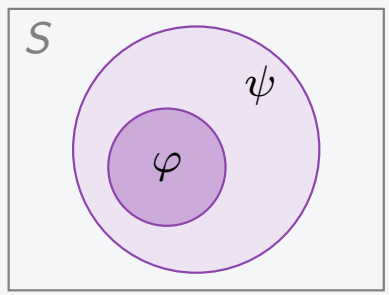
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In a picture



This is what makes an argument **valid**: the conclusion is true in every situation where the premises are. examples next

entailment $\models$

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A warm-up: checking entailment

The truth table

	P_a	P_b	$(P_a \vee P_b)$
s_1	**1**	1	**1**
s_2	**1**	0	**1**
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Question 1: $P_a \models (P_a \vee P_b)$ **?**

look only at the rows where the *premise* P_a is 1:

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rows where P_a is 1

Question 1: $P_a \models (P_a \vee P_b)$ **?**

look only at the rows where the *premise* P_a is 1:

is the conclusion 1? **yes**

$\{s_1, s_2\} \subseteq \{s_1, s_2, s_3\}$ ✓

A warm-up: checking entailment

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s_4	0	0	0	

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look only at the rows where the *premise* $(P_a \vee P_b)$ is 1:

is the conclusion 1? **no**: s_3

$\{s_1, s_2, s_3\} \not\subseteq \{s_1, s_2\}$ ✗

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The truth table

	P_a	P_b	$(P_a \vee P_b)$	
s_1	**1**	1	**1**	✓
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s_3 is a **counterexample**: the premise is 1, the conclusion is 0.

one counterexample is enough to show an entailment fails (lecture 3!)

Back to Mary: is the argument valid?

The argument

$$P_a \rightarrow (P_l \vee \neg P_f), \quad P_a \quad \therefore \quad P_l \vee \neg P_f$$

P_a: Mary can access the Internet P_l: Mary is a linguistics major

P_f: Mary is a freshman

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Join the premises with $\wedge$: one formula entails another

$$\underbrace{((P_a \rightarrow (P_l \vee \neg P_f)) \wedge P_a)}_{\text{the premises}} \models \underbrace{(P_l \vee \neg P_f)}_{\text{the conclusion}} \quad ?$$

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$$\underbrace{((P_a \to (P_l \vee \neg P_f)) \wedge P_a)}_{\text{the premises}} \models \underbrace{(P_l \vee \neg P_f)}_{\text{the conclusion}} \quad ?$$

By definition: $[\![\text{premises}]\!] \subseteq [\![\text{conclusion}]\!]$?

is the conclusion 1 in every row where the premises are 1? ($2^3 = 8$ rows)

Checking Mary's argument with a truth table

Bottom-up, as always

	P_a	P_l	P_f				
s_1	1	1	1				
s_2	1	1	0				
s_3	1	0	1				
s_4	1	0	0				
s_5	0	1	1				
s_6	0	1	0				
s_7	0	0	1				
s_8	0	0	0				

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	P_a	P_l	P_f	$\neg P_f$			
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	P_a	P_l	P_f	$\neg P_f$	conclusion		
s_1	1	1	1	0	**1**		
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$[\![\text{premises}]\!] = \{s_1, s_2, s_4\}$

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The argument is **valid**: premises $\models$ conclusion.

it follows from the *form* alone, whatever the facts about Mary are

We can now compute inferences!

An algorithm for entailment: does $\varphi \models \psi$?

1. list every situation: $S = \{s_1, \ldots, s_{2^n}\}$ all letters in φ and ψ
2. compute $[\![\varphi]\!]$: in each row, evaluate φ up its tree
3. compute $[\![\psi]\!]$ the same way
4. check: is $[\![\varphi]\!] \subseteq [\![\psi]\!]$?

   yes: $\varphi \models \psi$ **no**: any $s \in [\![\varphi]\!] \setminus [\![\psi]\!]$ is a counterexample

always gives an answer: there are only finitely many rows (but 2^n grows *very* fast!)

More notions, all defined with sets

Contingent

$\emptyset \neq [\![\varphi]\!] \neq S$

sometimes true, sometimes false

e.g., P_a, $(P_a \rightarrow P_b)$

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φ is *at least as* informative as ψ

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with $\subset$: *strictly* more informative

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Independent

all four sets are non-empty:

$[\![\varphi]\!] \cap [\![\psi]\!]$, $[\![\varphi]\!] \cap [\![\neg\psi]\!]$,
$[\![\neg\varphi]\!] \cap [\![\psi]\!]$, $[\![\neg\varphi]\!] \cap [\![\neg\psi]\!]$

knowing one tells you nothing about the other

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If φ is a tautology, then $\neg\varphi$ is unsatisfiable.

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$[\![\neg\varphi]\!] = S \setminus [\![\varphi]\!] = S \setminus S = \emptyset$ ■

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e.g., $(P_a \wedge P_b)$ is strictly more informative than P_a: $\{s_1\} \subset \{s_1, s_2\}$
P_a and P_b are independent: each of $s_1, \ldots, s_4$ is one combination

How our core notions are related

Proposition

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(1) $\varphi \models \psi$ $\rightsquigarrow$ truth tables

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in practice, each view leads to a different kind of algorithm

How many rows do we need in our truth table?

An algorithm for entailment: does $\varphi \models \psi$?

1. list every situation: $S = \{s_1, \ldots, s_{2^n}\}$ all letters in φ and ψ
2. compute $[\![\varphi]\!]$: in each row, evaluate φ up its tree
3. compute $[\![\psi]\!]$ the same way
4. check: is $[\![\varphi]\!] \subseteq [\![\psi]\!]$?

   yes: $\varphi \models \psi$ **no**: any $s \in [\![\varphi]\!] \setminus [\![\psi]\!]$ is a counterexample

always gives an answer: there are only finitely many rows (but 2^n grows *very* fast!)

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60	$\approx 10^{18}$	36 years
70	$\approx 10^{21}$	36 thousand years
80	$\approx 10^{24}$	36 million years
90	$\approx 10^{27}$	36 billion years
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$$2^{100} \text{ rows}$$

1 267 650 600 228 229 401 496 703 205 376

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In practice, our algorithm is not feasible

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A SAT solver *searches* for a row that makes the formula true, skipping huge parts of the table.

no solution: valid a solution: a counterexample

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Next time: We will be talking about extending our logic to have more constructs, **first-order logic**.